
NASH BARGAINING PREFERENCE OPTIMIZATION

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Paper under double-blind review

ABSTRACT

Multi-objective alignment requires both a representation of preferences and a rule for selecting a compromise. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective through a regularized preference game and applies Nash bargaining to improvement over a common reference policy. A constrained KL-proximal formulation handles zero surplus at the reference; a prompt-separable finite-pool solver accounts for adaptive opponents; and pairwise log-ratio regression transfers its target to an LLM without evaluating a language-model normalizer. Exact population updates admit an $O(1/T)$ last-iterate guarantee. Controlled experiments support adaptive game values as preference nontransitivity increases. In an 8B model, three policy seeds reproduce positive teacher-defined surpluses on SafeRLHF, but do not resolve a difference between Nash and mass-matched utilitarian aggregation. We separate this focused comparison from broader instruction-following, capability, and empirical trade-off evaluations, whose unmeasured entries remain explicitly marked. The current evidence does not establish an aggregation advantage or empirical Pareto optimality.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) commonly learns scalar rewards from pairwise comparisons (Christiano et al., 2017; Ouyang et al., 2022). Direct Preference Optimization removes the separate reward model but retains a Bradley–Terry (BT) preference model (Rafailov et al., 2023; Bradley & Terry, 1952). For a fixed prompt and objective, BT represents population probabilities as $P_k(y \succ z \mid x) = \sigma(r_k(x, y) - r_k(x, z))$. This permits noisy, contradictory sampled labels, but cannot represent a cyclic strict majority relation.

For example, under one fixed rubric, suppose $P_k(A \succ B) = P_k(B \succ C) = P_k(C \succ A) = 0.75$. No scalar scores can represent these three probabilities. This hypothetical within-rubric cycle differs from comparing relevance on one edge and safety on another. Whether such misspecification remains after defining separate objectives is an empirical question; sparse human comparison graphs and scalar-rating-induced labels do not establish it.

Nash Learning from Human Feedback (NLHF) retains general pairwise payoffs, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Multiple objectives additionally require a compromise rule. Existing methods use scalarization, constraints, policy interpolation, preference conditioning, or Blackwell criteria (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024; Agnihotri et al., 2026; Zhang et al., 2026).

We use classical Nash bargaining with equal bargaining weight for the specified objectives (Nash, 1950). It maximizes their surplus product over a common fallback without user-specified scalarization weights. Equal treatment is a choice: duplicating or splitting an objective changes its influence, and the fallback affects the result. Prior work already studies disagreement-adjusted Nash welfare over learned party rewards and bargaining over task-gradient improvements (Zhong et al., 2026; Navon et al., 2022).

Nash Bargaining Preference Optimization (NBPO) instead bargains over reference-relative *policy game values*. Each objective evaluates the learner against an adaptive, KL-regularized comparator. Improvement means a larger robustness-adjusted game value, not necessarily a direct win over

the reference. Population bargaining guarantees concern these specified utilities and do not imply per-prompt improvement.

The technical difficulty is the coupled proximal problem: surplus is zero at the reference, and each opponent depends on the policy being optimized. Pairwise normalizer cancellation and regression fitting are inherited tools (Zhang et al., 2025; 2026). Our optimizer derives the surplus dual, solves its prompt-separable finite-pool subproblems directly, and then fits their target to an LLM. This replaces an undamped best-response alternation that can be unstable even when its fixed point is optimal. Numerical target accuracy, neural realization, and independent policy quality are evaluated separately.

Our contributions are:

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, separating within-objective preference representation from across-objective compromise selection.
- We derive a practical optimizer that handles the zero-surplus boundary and policy-dependent opponents. For each dual iterate, the sampled finite-pool problem separates into small strictly concave prompt-wise programs; their target policy is expressed for the LLM as a pairwise log-ratio regression objective, without fitting a scalar reward or evaluating a language-model partition function. We evaluate both held-out target fit and downstream generation.
- We prove $O(1/T)$ last-iterate convergence for exact population proximal updates. Feasibility-preserving controlled experiments isolate the benefit of adaptive game values under increasing nontransitivity, while a human-labeled SAFERLHF panel evaluates neural transfer, objective-wise outcomes, and capability retention. Released natural datasets are not treated as evidence of within-objective cycles unless their comparison graphs identify such cycles.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y | x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z | x) \in [0, 1]$ satisfying

$$P_k(y \succ z | x) + P_k(z \succ y | x) = 1, \quad P_k(y \succ y | x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z | x) = P_k(y \succ z | x) - \frac{1}{2}, \quad \Delta_k(y, z | x) = -\Delta_k(z, y | x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot | x), z \sim \nu(\cdot | x)} \Delta_k(y, z | x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex feasible set that is downward-closed within $\{v : v \geq d\}$ and contains a strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G} : s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective’s pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k,\beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x))\}. \quad (6)$$

The comparator minimizes the learner’s pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z | x) := \mathbb{E}_{y \sim \pi(\cdot | x)} \Delta_k(y, z | x) \quad (7)$$

be the learner’s expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k,\pi}^*(z | x) = \frac{\pi_{\text{ref}}(z | x) \exp\{-m_k^\pi(z | x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot | x)} \exp\{-m_k^\pi(z' | x)/\beta_k\}}, \quad (8)$$

$$V_{k,\beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \pi_{\text{ref}}(\cdot | x)} \exp\{-m_k^\pi(z | x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k,β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_{\pi} V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x), \quad Q_k^\pi(y | x) := \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^\pi(y | x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y | x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

This is a substantive feasibility assumption. A finite-pool audit and failure outcomes are specified in Appendix A.5; the constrained formulation cannot make conflicting objectives jointly improvable. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_\beta(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_\beta := \{v \in \mathbb{R}^K : d \leq v \leq V_\beta(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_\beta$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_\beta, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. These properties concern population game values. Scaling primitive margins also requires scaling β_k ; arbitrary recalibration of judge probabilities is not covered (Appendix A.3). Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective is undefined at initialization. Second, every game value contains an opponent that depends on the learner. Third, the population policy update must be realized by an autoregressive model. We handle these issues with a constrained proximal problem, a prompt-separable finite-pool inner solve, and neural log-ratio projection.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_\pi F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^\pi(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$.

The dual is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}, \quad (17)$$

where $\lambda = (\lambda_1, \dots, \lambda_K) \in \mathbb{R}_{++}^K$ is the vector of Lagrange multipliers. For fixed λ , let

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

The KL term makes the maximizing policy $\pi_t^*(\lambda)$ unique. Danskin’s theorem (Danskin, 1967) then allows us to differentiate $\phi_t(\lambda)$ while holding this maximizing policy fixed, yielding

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the inverse-surplus weighting in (15) without selecting objective weights in advance.

5.2 DIRECT FINITE-POOL INNER SOLVE AND NEURAL PROJECTION

At outer stage t , draw learner occurrences $\mathcal{Y}_x = \{y_{x,i}\}_{i=1}^{n_x}$ from π_t and comparator occurrences $\mathcal{Z}_x = \{z_{x,j}\}_{j=1}^{m_x}$ from π_{ref} . Write $\hat{A}_{k,x}[i,j] = \hat{\Delta}_k(y_{x,i}, z_{x,j} \mid x)$. Retaining duplicate draws with multiplicity gives empirical base measures $p_{t,x}(i) = 1/n_x$ and $\mu_x(j) = 1/m_x$. These are importance-sampling occurrence weights, not normalized LM likelihoods. For $p \in \Delta^{n_x}$, define

$$\hat{m}_{k,x}(p)_j := \sum_{i=1}^{n_x} p_i \hat{A}_{k,x}[i,j], \quad (21)$$

$$\hat{V}_{k,x}(p) := -\beta_k \log \sum_{j=1}^{m_x} \mu_{x,j} \exp\{-\hat{m}_{k,x}(p)_j / \beta_k\}. \quad (22)$$

The fixed disagreement estimate $\hat{d}_k$ comes from a reference–reference tensor, not an assumed zero. Let $\hat{\mathcal{D}}$ be the empirical prompt distribution and set $\hat{s}_k(p) = \mathbb{E}_{x \sim \hat{\mathcal{D}}} \hat{V}_{k,x}(p_x) - \hat{d}_k$. The shared multipliers aggregate these prompt averages; there is no separate Nash bargain at each prompt.

For fixed λ , the empirical counterpart of (18) separates across prompts:

$$p_{t,x}^*(\lambda) \in \arg \max_{p \in \Delta^{n_x}} \left\{ \sum_{k=1}^K \lambda_k \hat{V}_{k,x}(p) - \frac{1}{\eta_t} \text{KL}(p \| p_{t,x}) \right\}. \quad (23)$$

Each problem is strictly concave because of the KL term. The opponent is analytically eliminated inside $\hat{V}$, so direct optimization accounts for its dependence on p . We solve these small programs to numerical tolerance, then solve the shared dual condition $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$ in $\log \lambda$. A computational box $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ requires reporting both unprojected stationarity and bound activity; projected stationarity alone does not certify the original problem. Appendix A.5 gives the residual checks and an explicit instability example for undamped policy–opponent alternation.

Once the dual converges, the finite-pool target defines the pairwise log-ratio change

$$\hat{h}_t^*(x, i, i') := \log \frac{p_{t,x}^*(i)}{p_{t,x}(i)} - \log \frac{p_{t,x}^*(i')}{p_{t,x}(i')}. \quad (24)$$

For a neural policy π_θ , let

$$h_t(\pi_\theta; x, y, y') := \log \frac{\pi_\theta(y \mid x)}{\pi_t(y \mid x)} - \log \frac{\pi_\theta(y' \mid x)}{\pi_t(y' \mid x)}. \quad (25)$$

We project the finite-pool target into the LLM with

$$\mathcal{L}_{\text{reg}}^{(t)}(\theta) = \mathbb{E}_{\substack{x \sim \hat{\mathcal{D}} \\ (i,i') \sim \xi_x}} \left[\left(h_t(\pi_\theta; x, y_{x,i}, y_{x,i'}) - \hat{h}_t^*(x, i, i') \right)^2 \right]. \quad (26)$$

The target is frozen during fitting; the square is inside the expectation. Complete pair coverage identifies only the reweighted sampled support, not unseen responses. At the inner optimum,

$$\hat{h}_t^*(x, i, i') = \eta_t \sum_{k=1}^K \lambda_k \left(\hat{Q}_{k,x}^*(i) - \hat{Q}_{k,x}^*(i') \right), \quad (27)$$

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , pairwise models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T .

```
1: Set  $\pi_0 \leftarrow \pi_{\text{ref}}$ ; construct fixed reference pools and  $\hat{d}_{1:K}$ .
2: for  $t = 0, \dots, T - 1$  do
3:   Draw learner pools from  $\pi_t$ , construct cross tensors, initialize  $\lambda > 0$ .
4:   repeat
5:     Solve (23) independently for every prompt.
6:     Update shared  $\lambda$  using  $\hat{s}_k - 1/\lambda_k$ ; check residuals and bounds.
7:   until unprojected KKT tolerance is met or the numerical budget is exhausted
8:   On uncensored output, audit feasibility, retain  $\pi_t$ , and return its failure status.
9:   Construct  $\hat{h}_t^*$  by (24); verify (27).
10:  Fit one neural candidate  $\tilde{\pi}_{t+1}$  using (26).
11:  Measure post-fit diagnostics on development data; accept only if declared checks pass.
12:  Set  $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$  if accepted, otherwise retain  $\pi_t$  and record rejection.
13: end for
14: return  $\pi_T$ .
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where $\hat{Q}_{k,x}^* = \hat{A}_{k,x} \nu_{k,x}(p_{t,x}^*)$. This identity checks the solver target independently. Pairwise probabilities may come from direct human comparisons or a general pairwise preference model; NBPO never requires them to decompose into scalar response rewards.

Neural pool diagnostics use $\tilde{p}_{\theta,x}(i) \propto p_{t,x}(i) \pi_{\theta}(y_{x,i} \mid x) / \pi_t(y_{x,i} \mid x)$, not normalized raw LM likelihoods. Appendix E compares regression with weighted behavior cloning of the same target.

Acceptance is empirical, uses development rather than final-test data, and provides no population certificate. Appendix A.5 defines failure reporting. Historical runs require artifact verification of this execution contract; all reported LLM runs use $T = 1$.

Last-iterate convergence. The following Bregman proximal-point guarantee concerns exact population solves, not finite-pool fitting or empirical acceptance; T counts exact solves rather than neural updates. Under Assumption 1, with $\pi_0 = \pi_{\text{ref}}$ and (16) solved exactly at every outer iteration with arbitrary $\eta_t > 0$, the Nash welfare is monotone, $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and $F(\pi^*) - F(\pi_T) \leq D(\pi^* \parallel \pi_{\text{ref}}) / \sum_{t < T} \eta_t$: a constant proximal step gives an $O(1/T)$ final-iterate gap. Theorem 1 and its proof are in Appendix A.6.

6 EXPERIMENTS

Our primary empirical questions are whether 8B policy optimization improves multiple objectives and how those gains relate to general capability. We organize the main comparisons around generated-policy outcomes, as in multi-objective alignment and NLHF (Agnihotri et al., 2026; Zhang et al., 2026; Wu et al., 2025). The broader instruction experiment below is specified but unmeasured; the completed three-seed SafeRLHF comparison remains visible.

6.1 GENERAL-INSTRUCTION ALIGNMENT AT 8B SCALE

Protocol. The planned main panel uses Llama-3.1-8B-Instruct and four UltraFeedback attributes: instruction-following, truthfulness, honesty, and helpfulness (Cui et al., 2024). Preference-model training uses released GPT-4 ratings; policy optimization uses newly sampled responses to disjoint prompts. The target split is 20k/2k preference-model train/dev, 10k/1k policy train/dev, and 2k final evaluation. Source filtering excludes benchmark-derived prompts; a predeclared 5k policy-training fallback applies if fewer than 35k eligible unique prompts remain. Appendix C.1 specifies the new four-head preference model, data processing, and update budget. These are proposed settings, not completed runs.

Objective trade-offs. Canonical NBPO is compared with mass-matched game-utilitarian, game-maxmin, actual PROSPER and MOPO adaptations, and scalarized DPO. Four specialist weight vectors, the uniform vector, and two prespecified interior vectors provide competing compromises.

Table 1: General-instruction 8B comparison template. IF, TF, Hon, and Help are independent local-judge win rates against the common base under four attribute rubrics; $\min \Delta$ is the smallest attribute-wise difference from 0.5. Planned training sizes and seed targets are not completed measurements. Dashes are unmeasured. The DPO row uses uniform weights; the figure retains every scalarization setting.

Method	N_{train}	Seeds	N_{eval}	IF	Truth	Honesty	Help	$\min_k \Delta_k$
Base	0	—	—	—	—	—	—	—
NBPO	10000	—	—	—	—	—	—	—
Game-utilitarian	10000	—	—	—	—	—	—	—
Game-maxmin	10000	—	—	—	—	—	—	—
PROSPER (adapt.)	10000	—	—	—	—	—	—	—
MOPO (adapt.)	10000	—	—	—	—	—	—	—
Scalarized DPO	10000	—	—	—	—	—	—	—

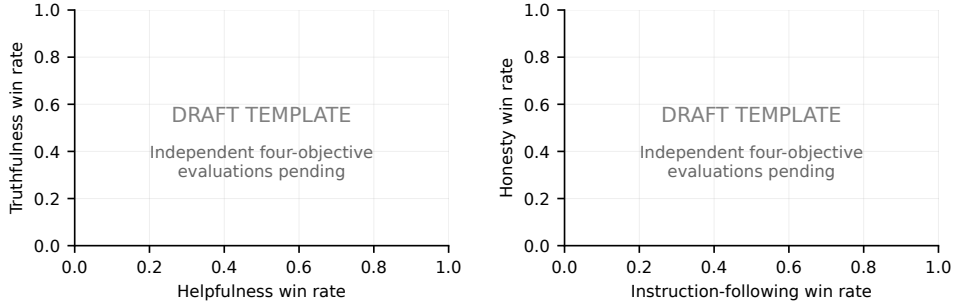


Figure 1: Template for the independent four-objective comparison. Left: helpfulness versus truthfulness; right: instruction-following versus honesty. No measurements have been supplied. Filled versions show measured policy points and uncertainty; two-dimensional projections do not determine four-dimensional dominance.

All methods share the base, prompt split, and declared feedback budget. Independent attribute evaluations are separated from training-teacher game surpluses. Nondominance is assessed in the full four-dimensional vector; Figure 1 displays two projections, not a certificate of global Pareto optimality.

6.2 GENERAL CAPABILITY AND SAFETY

Table 2 reports instruction following, reasoning, safety, benign refusal, and chat preference under common evaluation protocols. Alpaca and Arena columns are local proxies, not official leaderboard scores. Existing SafeRLHF runs are kept separate from the new instruction panel. Across seeds 42/43/44, their reported IFEval values average 0.748 for Nash and 0.745 for utilitarian, versus 0.752 for the base; these differences do not establish equivalence. Method-specific aggregates for the other metrics await the per-seed artifacts. No policy is selected using final-benchmark scores.

6.3 COMPLETED HELPFULNESS–SAFETY COMPARISON

SafeRLHF provides human helpfulness and harmlessness comparisons on the same response pairs and enabled matched GPM/BT validation. It is a focused safety panel rather than a requirement of NBPO. The completed comparison uses 2,000 policy-training prompts, 500 development prompts, 1,000 held-out prompts, and the fixed 250-update regression recipe. Both Nash and mass-matched utilitarian attain positive teacher-defined aggregate surplus across three policy seeds, without a resolved Nash advantage (Table 3). Their paired objective intervals include zero; this is not an equivalence result.

The earlier schedule comparison and pool diagnostics justify fixing the short recipe and are retained in Appendix E. They do not substitute for policy-quality comparisons. Teacher truncation affects 26,790/84,000 candidate occurrences, motivating a supported longer-context teacher for the new instruction panel. Controlled nontransitivity experiments separately support adaptive-game represen-

Table 2: 8B capability and safety template. UF rows are unmeasured. SafeRLHF IFEval means and sample standard deviations are computed from three reported seed values; the source metric variant awaits verification. HB is HarmBench harmful rate; XS is XSTest benign refusal, both lower is better. Alpaca/AH are local proxy win rates. Per-seed values and historical snapshots are in Appendix F.

Method	IFEval	GSM8K	MMLU	HB↓	XS↓	Alpaca	AH-hard	AH-creat.
<i>UF-4: planned; no measurements</i>								
Base	—	—	—	—	—	—	—	—
NBPO	—	—	—	—	—	—	—	—
Game-utilitarian	—	—	—	—	—	—	—	—
Game-maxmin	—	—	—	—	—	—	—	—
PROSPER (adapt.)	—	—	—	—	—	—	—	—
MOPO (adapt.)	—	—	—	—	—	—	—	—
Scalarized DPO	—	—	—	—	—	—	—	—
<i>SafeRLHF-2k: reported three-seed study</i>								
Base	0.752	0.867	—	0.281	—	0.500	—	—
NBPO	0.748 ± 0.013	—	—	—	—	—	—	—
Game-utilitarian	0.745 ± 0.025	—	—	—	—	—	—	—

Table 3: Completed SafeRLHF comparison, three policy seeds per method. Entries reproduce reported means and spreads; the spread convention and minimum/averaging order require artifact verification. Surplus uses the training teachers and does not certify population individual rationality.

Family	Seeds	nMSE↓	Sign↑	$\min_k s_k \uparrow$
NBPO-Nash	3	0.6845 ± 0.0048	0.7009 ± 0.0016	0.1215 ± 0.0028
Utilitarian (L1-matched)	3	0.6737 ± 0.0242	0.7040 ± 0.0058	0.1220 ± 0.0029

tations: at the strongest perturbation, normalized worst surplus is 0.931 for NBPO versus -0.241 for fixed-reference Nash; game-utilitarian also remains stable (Appendix B). Thus those results isolate representation, not Nash superiority. The matched natural-language adaptive-versus-fixed-reference comparison remains pending; Nash versus utilitarian isolates aggregation.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026). NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024). MaxMin-RLHF and RMOD optimize weakest-objective criteria (Chakraborty et al., 2024; Son et al., 2026). MOPO uses pairwise preferences without point-wise rewards, selecting a primary objective and lower-bound constraints with a lagged reference in its practical algorithm (Agnihotri et al., 2026). RACO resolves conflicts between DPO-style objective gradients (Chen et al., 2026). These approaches differ from bargaining over reference-relative robust policy values.

Social choice and bargaining. The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2026); Nash bargaining also selects immediate task-gradient improvements in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

NBPO combines adaptive preference-game values with Nash bargaining over reference-relative improvements. A constrained proximal formulation yields a direct finite-pool solver and an $O(1/T)$ exact-population guarantee. Controlled experiments support adaptive game representations. Three policy seeds reproduce teacher-defined gains on SafeRLHF without a resolved advantage over mass-matched utilitarian aggregation. Broader instruction and independent objective comparisons remain unmeasured. Teacher truncation, refusal-concentrated gains, and development-informed schedules limit the current evidence; empirical Pareto optimality is not established. Joint improvement need not be feasible, and population game values provide no per-prompt safety guarantee.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 distinguishes the population proximal problem, the numerically solved finite-pool inner problem, and its neural projection. Appendix A contains the dual and convergence proofs; Appendices B–F specify constructions, supervision, hyperparameters, diagnostics, and evaluation protocols. The submission package should include candidate pools, preference tensors, dual traces, solver residuals, per-seed outputs, and the code and configurations required to reproduce the reported results.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin’s theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (28)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_{\nu} J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

Game-value improvement need not imply a direct reference win. For one prompt, consider

$$A = \begin{pmatrix} 0 & .5 & -.5 \\ -.5 & 0 & .5 \\ .5 & -.5 & 0 \end{pmatrix}, \quad \mu = (.8, .1, .1), \quad p = (.25, .43, .32), \quad \beta = .25.$$

Both policies have full support. Direct evaluation gives $V_\beta(\mu) = -.0517901$ and $V_\beta(p) = -.0417293$, hence $s(p) = .0100608 > 0$. Yet $p^\top A\mu = -.0385$, so the direct reference win probability is .4615. Robustness-adjusted value and direct wins answer different questions even under the same judge.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k,β_k} is continuous, $V_\beta(\pi)$ is bounded. The definition of $\mathcal{G}_\beta$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_\beta$, choosing policies π_n with $d \leq v_n \leq V_\beta(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure within $\{v : v \geq d\}$ is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_\beta$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_\beta(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_\beta(\pi_1) + (1 - \alpha)V_\beta(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_\beta$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_\beta$ with $v > d$, some policy π satisfies $v \leq V_\beta(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k,\beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_\beta(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_\beta, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatewise monotonicity implies Pareto optimality. Strict concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

Table 4: Guarantees of the stated population rules under joint improvement. A “no” means the property need not hold for every optimizer without an additional tie-break. This is not an exhaustive comparison of bargaining solutions.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

Scope of invariance and alternatives. Affine invariance transforms completed utilities and disagreement values together. At the primitive level, $A'_k = c_k A_k$ requires $\beta'_k = c_k \beta_k$ to preserve the opponent and obtain $V'_k = c_k V_k$. With fixed temperature instead, $V_{cA,\beta} = c V_{A,\beta/c}$. If scaled margins leave $[-1/2, 1/2]$, the transformation changes payoff units rather than defining a new probability oracle. Arbitrary judge recalibration is not covered.

Exact dual multipliers transform as $\lambda'_k = \lambda_k/c_k$, leaving $\lambda'_k Q'_k = \lambda_k Q_k$ and the pair target unchanged. Matching projected-gradient trajectories further requires scaling initial multipliers and coordinate bounds by $1/c_k$, any diagnostic surplus thresholds by c_k , and coordinate step sizes by $1/c_k^2$. Target signs alone do not verify these stronger identities. We do not floor surpluses to make infeasible targets positive.

A Pareto-refining tie-break can remove the dominated-optimum failure of raw maxmin. Relative-improvement bargaining provides another scale-invariant choice but requires objective-wise ideal values and, in general, an appropriate refinement (Kalai & Smorodinsky, 1975; Han et al., 2026). These alternatives do not make Nash uniquely admissible. Here the bargaining parties are objective rubrics: duplicating a rubric duplicates its log-surplus term. The fixed reference also plays two distinct roles, disagreement policy and opponent anchor; changing both changes the utilities as well as disagreement.

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

The positive guarantees in Table 4 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maxmin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maxmin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

For the uniform reference,

$$\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$$

and hence

$$d = (-1/3, -1/12).$$

First, joint improvement is feasible. The policy

$$\bar{\pi} = (1/4, 1/4, 1/2)$$

is only a feasibility witness; it gives

$$\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$$

Therefore

$$V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$$

In contrast, solving the equal-weight utilitarian problem $\max_{\pi} V_1(\pi) + V_2(\pi)$ gives

$$\pi_U = (0, 1/5, 4/5).$$

For this policy,

$$\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$$

so

$$V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$$

Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

Absolute-maxmin failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again,

$$d = (-1/3, -1/12).$$

Joint improvement is feasible, as witnessed by

$$\bar{\pi} = (1/3, 7/15, 1/5),$$

for which

$$V(\bar{\pi}) = (-4/15, -1/60)$$

and therefore

$$s(\bar{\pi}) = (1/15, 1/15) > 0.$$

Absolute maxmin instead solves

$$\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$$

whose linear-program solution is

$$\pi_{AM} = (0, 4/5, 1/5).$$

For this policy,

$$\pi_{AM}^\top \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{AM}^\top \Delta_2 = (-1/10, -1/10, 2/5),$$

so

$$V(\pi_{AM}) = (-1/10, -1/10)$$

and

$$s(\pi_{AM}) = (7/30, -1/60).$$

Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed $1/6$.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at $1/6$. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION, DIRECT SOLVE, AND NEURAL PROJECTION

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin’s theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater’s condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Dual step used as a fallback. Stationarity $\hat{s}_k(p_t^*(\lambda)) = 1/\lambda_k$ can also be approached by projected subgradient *descent* on the dual — the update below subtracts the dual gradient, and the dual of the concave inner problem is minimized over λ —

$$\lambda^{(m+1)} = \Pi_\Lambda \left[\lambda^{(m)} - \gamma_m \left(\hat{s} \left(p_t^* \left(\lambda^{(m)} \right) \right) - (\lambda^{(m)})^{-1} \right) \right], \quad (29)$$

where $\gamma_m > 0$ is the dual step size, Π_Λ is Euclidean projection onto $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$ with $0 < \lambda_{\min} < \lambda_{\max}$, and the inverse is elementwise. On the pools reported here this iteration stalls at a projected KKT residual near 10^{-2} after 3000 steps, whereas the root solve reaches 10^{-9} in tens of inner solves; no reported result uses (29).

What the projected certificate certifies. The projected residual concerns the box-constrained dual. Original stationarity additionally requires $\|\hat{s} - \lambda^{-1}\|_\infty$ to meet tolerance; active bounds cannot be ignored merely because projection removes the residual. We record each bound hit and classify a solve with unresolved unprojected stationarity as uncertified. Interior numerical solutions with small residuals support the finite problem only. The supplied SafeRLHF report gives $\Lambda = [10^{-3}, 10^3]^2$, $\lambda = (6.33, 6.19)$, and independently recomputed unprojected residual 6.7×10^{-11} . These are teacher diagnostics, not neural-policy certificates.

Finite-pool optimality and neural projection. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (30)$$

The direct finite-pool program (23) is the sampled analogue of this strictly concave problem. Its normalized optimizer determines all finite-pool log-probability differences, so (24) is identified without computing a language-model normalizer. Equation (27) is the corresponding finite-pool KKT identity and is checked numerically before an artifact is accepted. If the positive-probability pair graph connects all sampled occurrences, zero regression error implies

$$\tilde{p}_{\theta, x}(i) = \frac{p_{t, x}(i) \pi_{\theta}(y_{x, i} | x) / \pi_t(y_{x, i} | x)}{\sum_j p_{t, x}(j) \pi_{\theta}(y_{x, j} | x) / \pi_t(y_{x, j} | x)} = p_{t, x}^*(i).$$

This identifies the importance-reweighted occurrence measure only. For example, under a uniform three-response anchor, fitting only pair (1, 2) with target zero permits every neural distribution $(a, a, 1 - 2a)$, $0 < a < 1/2$. Unseen-response mass remains unconstrained. Resampling a new learner pool at each outer stage makes the empirical base measure track π_t ; reusing a pool from another proposal instead requires explicit importance weights. The reference opponent anchor remains π_{ref} .

Why an optimal fixed point need not be reached. Use the cyclic matrix A above, reference and proximal center $\mu = (.34, .33, .33)$, $\beta = .25$, and fixed $\eta\lambda = 4$. Undamped best-response alternation applies

$$T(p) = \text{softmax}(\log \mu + 4A \text{softmax}(\log \mu + Ap/\beta)).$$

Its fixed point is $p^* = (.33618258, .32907171, .33474571)$, but both tangent-space Jacobian eigenvalues are approximately -1.333163 . Perturbations therefore alternate and expand locally. This is not excluded by improvement: $V(p^*) - V(\mu) \simeq 4.7194 \times 10^{-5} > 0$. In the supplied deterministic check, 100 iterations from μ leave $\|p - p^*\|_2 \simeq .27986$; a direct constrained solve of the same fixed-multiplier problem reaches inner stationarity below 10^{-9} . These are finite-game calculations, not LLM results or a complete Nash dual solve.

For general fixed λ , let $f_x(p; \lambda)$ denote the objective in (23) and let $\nu_k(p) = \text{softmax}(\log \mu_x - \hat{A}_{k, x}^\top p / \beta_k)$. On the simplex interior,

$$\begin{aligned} \nabla f_x(p; \lambda) &= \sum_k \lambda_k \hat{A}_{k, x} \nu_k(p) - \eta_t^{-1} (\log(p/p_{t, x}) + \mathbf{1}), \\ \nabla^2 f_x(p; \lambda) &= - \sum_k \frac{\lambda_k}{\beta_k} \hat{A}_{k, x} (\text{diag}(\nu_k) - \nu_k \nu_k^\top) \hat{A}_{k, x}^\top - \eta_t^{-1} \text{diag}(1/p) \prec 0. \end{aligned}$$

Thus the finite target is unique, and direct optimization includes opponent feedback through this curvature. Strict concavity does not certify an arbitrary numerical routine: simplex feasibility and inner/dual residuals still require measurement. The empirical tensor itself remains an approximation to population preferences; exponentiation, normalization, and log-sum-exp do not preserve unbiasedness of noisy margins.

Finite-pool feasibility and failure reporting. When diagnosing an uncertified solve, use the finite-pool program

$$\hat{\tau}^* = \max_{\{p_x\}, \tau} \tau \quad \text{s.t.} \quad \mathbb{E}_{x \sim \hat{\mathcal{D}}} \hat{V}_{k, x}(p_x) - \hat{d}_k \geq \tau \quad (\forall k), \quad p_x \in \Delta^{n_x}.$$

This is convex optimization with concave-superlevel constraints, not generally a linear program at $\beta_k > 0$. A verified positive feasible margin witnesses strict finite-pool improvement. Only a valid nonpositive upper bound certifies its absence; failure to find a witness is unresolved. A tolerance-bounded interval near zero means no resolved positive margin, not exact infeasibility. Independent learner and reference banks need not contain a categorical representation of the fallback, so finite-pool infeasibility is not population infeasibility.

For an analytic infeasible example, take two actions, $\mu = (1/2, 1/2)$, $A_1 = \begin{pmatrix} 0 & 1/2 \\ -1/2 & 0 \end{pmatrix}$ and $A_2 = -A_1$. For $p = (a, 1 - a)$ and any common $\beta > 0$, $s_1(p) = (a - 1/2)/2 = -s_2(p)$, hence $\tau^* = 0$. No clipping or inverse-surplus floor creates a strictly feasible Nash target.

The execution contract records feasible, certified-infeasible, numerical-unresolved, and neural-rejected outcomes separately. Uncertified solves retain the previous policy. This audit introduces no separate neural warm-start stage. Post-fit acceptance uses prespecified development checks, including empirical surplus signs, target fit, and numerical validity; reused monitoring prompts are validation data. False acceptance/rejection rates near the boundary remain an unmeasured audit, and generic bounded-mean confidence formulas do not automatically cover biased game-value estimates.

A.6 LAST-ITERATE CONVERGENCE

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (31)$$

Work throughout in the $\mathcal{D}$ -weighted geometry: pair functions of (x, y) by $\langle f, g \rangle_{\mathcal{D}} = \mathbb{E}_{x \sim \mathcal{D}} \sum_y f(x, y)g(x, y)$, and let h be the *prompt-wise* negative entropy $h(\pi)(x) = \sum_y \pi(y | x) \log \pi(y | x)$, whose gradient at (x, y) is $1 + \log \pi(y | x)$. Its Bregman divergence in that geometry is exactly $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$. The prompt expectation enters once, through the pairing, and is not also carried inside h . Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (32)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (32) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (33)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (35)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (36)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (35), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_{\text{ref}}). \quad (37)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (31).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (38)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because $\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (39)$$

Applying this inequality to $s = s(\pi_T)$ and using (31) proves (38). $\square$

B CONTROLLED BENCHMARK: CONSTRUCTION AND FULL RESULTS

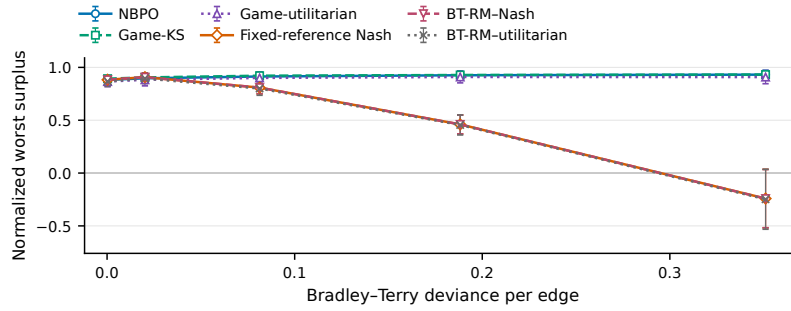


Figure 2: Controlled nontransitivity. Points show mean BT deviance and normalized worst surplus across five seeds; bars show one sample standard deviation. All six plotted methods converge. Full results and the legacy-map ablation are in Appendix B.

Feasibility-preserving construction. For every seed we form a transitive base matrix and an odd-cardinality circulant skew-symmetric perturbation. Zero row sums keep the uniform reference disagreement invariant, while bisection selects the cycle magnitude so that $\rho^* = \max_{\pi} \min_k s_k(\pi)$ stays close to a target margin. We reject invalid probabilities rather than clip matrix entries. The resulting panel increases BT deviance from approximately 2.7×10^{-4} to 0.351 while retaining positive joint improvement in every instance. All policies are evaluated under the same adaptive-game values, not their method-native scales.

Robustness across construction and temperature. The primary construction is checked against an independently generated projected skew-symmetric family, $C = N(R - R^T)N$ with N the symmetric projector annihilating both μ and a witness policy, and against a temperature sweep on that family. The trend replicates: adaptive-game rows stay flat while fixed-reference and scalar-reward rows fall. The effect *size* does not: the $\alpha = 1$ gap is +0.111 on the projected family against +1.172 on the circulant one, because the projected family’s BT deviance only reaches 0.174 against 0.351. We therefore claim robustness of the trend and do not attach the circulant effect size to it. Intervals in Table 7 are exact paired Student-*t* intervals over seeds, not bootstrap intervals.

C NATURAL SUPERVISION AND PREFERENCE MODELS

Dataset audit. Table 8 records the empirical role of each released dataset. ULTRAFEDBACK supplies complete per-completion ordinal attributes, so score-induced pairwise labels are transitive by construction. SAFERLHF supplies direct helpfulness and safety comparisons but almost exclusively isolated edges; zero observed triangles means the data cannot test within-objective cycles. It nevertheless contains substantial cross-objective disagreement and is therefore appropriate for the bargaining experiment.

Table 5: Full controlled results over five seeds under the common adaptive-game evaluator. Residuals are method-specific; the legacy map is a non-convergence ablation.

method	min surplus	sd	min s/ρ^*	sd	exploitability	inner resid.
$\alpha = 0.00, \rho^* = 0.0300, BT \text{ dev.} = 0.0003$						
NBPO direct	+0.0264	0.0011	0.881	0.037	0.0687	2×10^{-12}
Game-KS	+0.0268	0.0005	0.893	0.017	0.0685	10^{-9}
Game-util.	+0.0260	0.0014	0.866	0.048	0.0688	3×10^{-10}
Fixed-ref. Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-util.	+0.0260	0.0014	0.867	0.047	0.0689	0
legacy R -step [†]	+0.0252	0.0012	0.840	0.041	0.0675	5×10^{-10}
$\alpha = 0.25, \rho^* = 0.0300, BT \text{ dev.} = 0.0202$						
NBPO direct	+0.0269	0.0013	0.897	0.044	0.1091	10^{-7}
Game-KS	+0.0272	0.0007	0.905	0.025	0.1088	10^{-7}
Game-util.	+0.0265	0.0017	0.883	0.058	0.1094	4×10^{-8}
Fixed-ref. Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-util.	+0.0269	0.0015	0.895	0.049	0.1198	0
legacy R -step [†]	+0.0258	0.0014	0.860	0.046	0.1063	7×10^{-1}
$\alpha = 0.50, \rho^* = 0.0300, BT \text{ dev.} = 0.0813$						
NBPO direct	+0.0274	0.0013	0.914	0.044	0.1454	2×10^{-6}
Game-KS	+0.0276	0.0007	0.921	0.023	0.1450	2×10^{-7}
Game-util.	+0.0270	0.0017	0.900	0.057	0.1458	2×10^{-7}
Fixed-ref. Nash	+0.0243	0.0018	0.809	0.059	0.1945	0
BT-RM-Nash	+0.0243	0.0018	0.809	0.058	0.1945	0
BT-RM-util.	+0.0240	0.0019	0.800	0.064	0.1947	0
legacy R -step [†]	+0.0025	0.0044	0.082	0.147	0.1981	10×10^{-1}
$\alpha = 0.75, \rho^* = 0.0300, BT \text{ dev.} = 0.1882$						
NBPO direct	+0.0278	0.0013	0.926	0.042	0.1693	8×10^{-6}
Game-KS	+0.0278	0.0005	0.928	0.018	0.1692	8×10^{-7}
Game-util.	+0.0273	0.0017	0.909	0.056	0.1696	7×10^{-7}
Fixed-ref. Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-util.	+0.0137	0.0028	0.455	0.095	0.2724	0
legacy R -step [†]	-0.0453	0.0060	-1.511	0.201	0.2626	10×10^{-1}
$\alpha = 1.00, \rho^* = 0.0283, BT \text{ dev.} = 0.3510$						
NBPO direct	+0.0265	0.0033	0.931	0.041	0.1824	5×10^{-5}
Game-KS	+0.0264	0.0027	0.932	0.019	0.1825	3×10^{-6}
Game-util.	+0.0258	0.0037	0.908	0.063	0.1826	2×10^{-6}
Fixed-ref. Nash	-0.0063	0.0068	-0.241	0.277	0.3464	0
BT-RM-Nash	-0.0063	0.0068	-0.242	0.276	0.3464	0
BT-RM-util.	-0.0065	0.0070	-0.248	0.284	0.3465	0
legacy R -step [†]	-0.0949	0.0064	-3.373	0.406	0.3337	10^0

Table 6: Finite-pool solver audit at $\alpha = 1$ in the original controlled family. The direct practical solve recovers the positive-surplus proximal target; the deployed legacy map does not.

solution	min surplus	$/\rho^*$	inner residual	TV to exact prox.
Exact global Nash	+0.0158	+0.935	—	0.007
Exact proximal Nash	+0.0158	+0.936	—	0.000
NBPO direct inner solve	+0.0139	+0.821	2×10^{-6}	0.027
Legacy R -step map	-0.1851	-10.938	10^0	0.643
BT-RM-Nash	-0.1001	-5.918	0	0.495

Anti-symmetric general preference model. For objective k , a joint encoder produces $a_{\phi,k}(x, y, z)$ and we set

$$\ell_{\phi,k}(x, y, z) = \frac{1}{2} \{a_{\phi,k}(x, y, z) - a_{\phi,k}(x, z, y)\}, \quad P_{\phi,k}(y \succ z \mid x) = \sigma(\ell_{\phi,k}(x, y, z)).$$

This guarantees exact swap complementarity and self-ties without constraining the logit to a scalar reward difference. In the SafeRLHF diagnostic, both the joint model and the scalar BT baseline use RoBERTa-base so that encoder capacity and training cost are matched; this backbone choice is an experimental control rather than part of the NBPO method. Because the observed SafeRLHF graph contains no triangles, their near-null comparison is a calibration result rather than evidence that human preferences are transitive.

Table 7: Robustness across construction and temperature. Gaps in normalized worst surplus use paired Student- t 95% intervals across seeds. The temperature sweep uses the projected family.

cycle family	β	seeds	NBPO—FixedRef (t_{95})	NBPO—BT-RM (t_{95})	min ρ^*	converged
Circulant (primary)	0.25	5	+1.1724 [+0.8715, +1.4733]	+1.1733 [+0.8735, +1.4731]	0.0283	150/150
Projected skew-symmetric	0.25	5	+0.1114 [+0.0714, +0.1514]	+0.1115 [+0.0715, +0.1514]	0.0332	150/150
Projected skew-symmetric	0.10	3	+0.2381 [+0.0880, +0.3883]	+0.2385 [+0.0883, +0.3887]	0.0331	90/90
Projected skew-symmetric	0.50	3	+0.0448 [+0.0165, +0.0730]	+0.0449 [+0.0167, +0.0730]	0.0334	90/90

Table 8: Released-supervision audit. Zero observed human triangles means cyclicity is unidentifiable, not absent. Human conflict is cross-objective label disagreement on held-out pairs.

dataset	rows	prompts	responses	obs. triangles	human conflict	supervision	role
SAFERLHF	73,907	38,639	143,668	0	24.33%	direct pairwise	objective conflict
ULTRAFEEDBACK	63,967	63,967	4/prompt	n/a (score-induced)	n/a	ordinal attributes	transitive control

C.1 GENERAL-INSTRUCTION DATA AND PREFERENCE MODELS

The proposed UltraFeedback panel retains the UltraChat, ShareGPT, and Evol-Instruct sources, excludes TruthfulQA, FalseQA, and FLAN sources, and removes benchmark overlaps before splitting by normalized prompt identity. Target sizes are 20k/2k preference-model train/dev, 10k/1k policy train/dev, and 2k final evaluation. If filtering leaves 30k–35k eligible unique prompts (excluding 35k), policy training is reduced to 5k before training begins; fewer than 30k requires an explicitly revised data budget; exact retained counts and hashes must be reported. No final prompt is used for preference-model or policy selection.

Released GPT-4 ratings train four attribute heads. Valid ratings induce 0/1 pair labels, ties use 0.5, and missing labels are masked. Four completions produce six unordered pairs per prompt. These labels are transitive by construction and do not demonstrate human preference cycles. Policy-training prompts are disjoint and use new base-policy responses; no new paid judge API is required to reuse the released annotations.

The proposed GPM and scalar BT model share the 149M ModernBERT-base encoder with native 8192-token support (Warner et al., 2024). The GPM retains the antisymmetric construction above; BT uses differences of four scalar scores. They share supervision, one frozen preference-model seed, and prompt splits; policy-seed repeats do not measure teacher uncertainty. Initial fine-tuning settings are AdamW, learning rate 2×10^{-5} , batch 32 pairs, weight decay 0.01, warmup 0.06, and up to two epochs selected on preference-model development data. Lengths are bucketed, unsupported inputs are audited, and generated-policy evaluation reads the full response under a separate judge.

The proposed policy recipe retains full 8B fine-tuning, batch 32 pairs, and the existing peak learning-rate scale, with 1250 updates for 10k prompts or 625 for 5k. This fixes four sampled pair exposures per training prompt on average; it is not a claim that the budget is optimal. Each run uses its own cosine schedule and one outer stage. Proposal generation uses independent learner/reference pools of eight occurrences each. Reference–reference scoring and the shared tensor are cached across methods. With two encoder orders and four joint heads, 10k prompts require $2 \times 10^4(64 + 28) = 1.84$ million encoder passes; actual throughput, memory, and token counts determine the execution budget.

Aggregation across objectives and across prompts. NBPO applies $\sum_k \log(V_k(\pi) - d_k)$ to prompt-averaged objective values. In contrast, PROSPER’s regularized criterion has prompt-dependent weights (Zhang et al., 2026):

$$\min_{w: \mathcal{X} \rightarrow \Delta_K, \nu} \mathbb{E}_x \left[\sum_k w_k(x) J_{k,x}(\pi, \nu) + \beta \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)) \right],$$

where $J_{k,x}$ is the conditional pairwise payoff. Constant shifts from centered payoffs do not change its optimizer. Prompt-dependent weights cannot be replaced by a shared weight without changing this criterion. Prompt-direct optimization decomposes only the inner computation; NBPO’s λ remains shared across prompts.

Table 9: Held-out validation of checkpointed GPM and BT ensembles, seeds 41/42/43. Calibration uses validation data. Sparse comparison graphs cannot establish a cyclicity advantage.

model	objective	per-seed (mean of 3)			T	calibrated 3-seed ensemble				
		NLL	bal. acc	AUC		NLL	bal. acc	AUC	ECE	ens. sd
Anti-symmetric GPM	helpfulness	0.5543	0.7218	0.7999	1.282	0.5512	0.7257	0.8034	0.1228	0.0293
Anti-symmetric GPM	harmless.	0.5180	0.7556	0.8464	1.297	0.5131	0.7573	0.8512	0.1782	0.0357
Scalar BT	helpfulness	0.5560	0.7214	0.7974	1.270	0.5527	0.7250	0.8009	0.1220	0.0300
Scalar BT	harmless.	0.5212	0.7507	0.8431	1.265	0.5163	0.7544	0.8484	0.1779	0.0357

Table 10: Generated-pool pilot on 200 development prompts. Conflict and cycles are model predictions, not human-label measurements; cycles use a common comparator tournament.

geom.	dup. rate	entropy	ens. sd	GPM/BT sign	model conflict	pred. cycles	split-half r	ρ	sign agr.	dual evals	solve (s)
4+4	0.0073	0.6117	0.0429	0.9102	7.67%	0/800	0.9172	0.9132	0.8667	42	24.8
8+8	0.0167	0.6120	0.0414	0.9061	7.80%	10/11200	0.9242	0.9180	0.8925	30	18.4

To illustrate aggregation order alone, take two equally likely prompts. A policy with surplus vectors $(.2, 0)$ and $(0, .2)$ has average $(.1, .1)$; another with $(.08, .08)$ on both prompts has average $(.08, .08)$. Global Nash prefers the first, whereas the average prompt-wise minimum is 0 versus .08 and prefers the second. This schematic example omits opponent regularization and is not a numerical PROSPER result. We therefore distinguish a faithful PROSPER adaptation from global game-maxmin over NBPO values.

D TRAINING PROTOCOL AND BASELINES

The completed Nash/utilitarian comparison uses one outer stage and policy seeds 42/43/44. Earlier WBC and long-schedule diagnostics each use one policy seed. Comparisons within the repaired panel share the stated prompts, teacher, pools, and realization settings. Equal update counts do not imply equal FLOPs. Table 11 records the original recipe; resolved configurations and external-baseline results still require source verification.

E SOLVER AND NEURAL-REALIZATION DIAGNOSTICS

The direct solve concerns a finite pool at a declared tolerance. It does not make the preference model, sampling approximation, or neural fit exact. Solver consistency, held-out regression, and generated-policy evaluation are reported separately.

Alternative realization. For comparison with (26), weighted behavior cloning uses

$$\mathcal{L}_{\text{WBC}}^{(t)}(\theta) = \mathbb{E}_x \sum_{i=1}^{n_x} p_{t,x}^*(i) [-\log \pi_\theta(y_{x,i} | x)]. \quad (40)$$

Weights are detached and full-vocabulary response log-probabilities are summed over tokens. With IID reference-policy occurrences, this weighted likelihood approximates population forward-KL projection; it remains a finite-sample realization rather than an exact population update (Agnihotri et al., 2026). Averaging $\frac{n_x}{2} [p_i^* (-\log \pi_\theta(y_i | x)) + p_j^* (-\log \pi_\theta(y_j | x))]$ over all unordered pairs equals the prompt-wise objective. Uniform teacher weights need not produce zero gradient on a finite sampled dataset.

Metrics and selection history. Write $t = \hat{h}_t^*$ and $h = h_t(\pi_\theta)$. We report $\text{nMSE} = \mathbb{E}[(h - t)^2] / \mathbb{E}[t^2]$, sign agreement, Pearson and Spearman correlations. Expectations use the stated split and pair measure. The zero predictor has nMSE 1. The reported oracle rescaling diagnostic is

$$a^* = \frac{\mathbb{E}[ht]}{\mathbb{E}[h^2]}, \quad \text{nMSE}^* = \min_{a \in \mathbb{R}} \frac{\mathbb{E}[(ah - t)^2]}{\mathbb{E}[t^2]}.$$

Table 11: Reported study dimensions and announced repair settings. Optimizer, precision, and sampling settings must be reconciled with the execution manifest. This table concerns the completed SafeRLHF panel, not the planned UltraFeedback protocol.

Setting	Value
Base / adaptation	Llama-3.1-8B-Instruct / full fine-tuning
Train / dev / held-out prompts	2,000/500/1,000
Policy seeds / outer stages	42/43/44 (Nash/utilitarian); 1 stage
Learner / comparator occurrences	8/8; independent streams
Sampling	temperature 1, top- $p = 1$; preserve terminal tokens
Preference models	frozen three-seed objective ensembles
Inner solve / neural target	direct finite-pool / canonical log-ratio
Optimizer / learning rate	AdamW / 5×10^{-7}
Adam moments / epsilon	(0.9, 0.95) / 10^{-8}
Schedule / declared updates	cosine / 1750 or 250
Warmup ratio / weight decay	0.10/0
Global batch / pair rows	32 pairs / 56,000
Prompt / total sequence limit	1024/2048 tokens
Precision / gradient clip	bf16 forward, fp32 reduction / 1
Opponent temperatures / proximal η	— (resolved config required)
Reference cache / optimizer-state checks	— (execution report required)
Model / split / config revisions	— (immutable manifest required)

Table 12: Matched mechanism controls. External PROSPER and MOPO comparisons require their actual algorithms and separate tuning contracts; game-maxmin or WBC alone is not their reproduction.

method	within-objective representation	aggregation	changed component relative to NBPO
NBPO	adaptive regularized game	Nash	—
Fixed-reference Nash	fixed reference margin	Nash	removes adaptive opponent
BT-RM-Nash	scalar BT reward	Nash	imposes scalar reward representation
Game-utilitarian	adaptive regularized game	equal-weight sum	replaces Nash aggregation
BT-RM-utilitarian	scalar BT reward	equal-weight sum	scalar representation and aggregation control

It fits a scalar on the evaluation observations and can reverse sign. It is neither centered explained variance nor a deployable calibration result. A direction-preserving version restricts $a \geq 0$; signed coefficients and this additional metric require the raw moments. If $\mathbb{E}[h^2] = 0$, set $a^* = 0$ and the residual to 1. Mean response log-ratio is measured under the sampled occurrence distribution; it is not total pool probability or a full-policy KL certificate.

The 1750-update comparison was specified first. The 250-update schedules were chosen after inspecting development trajectories and declared before their runs. Each uses its own warmup and cosine horizon. A checkpoint at step 250 of the long run is therefore a different condition. Diagnostic thresholds were retained, but their use in development precludes describing the entire 2×2 as an untouched preregistered comparison.

Finite-pool distribution fidelity. For a candidate occurrence i , write $\Delta_i = \log \pi_\theta(y_i | x) - \log \pi_t(y_i | x)$ and reconstruct

$$\hat{p}_\theta(i | x) = \frac{p_{t,x}(i)e^{\Delta_i}}{\sum_j p_{t,x}(j)e^{\Delta_j}}.$$

This self-normalized importance distribution is evaluated against the solved target by forward KL on the fixed sampled support. It does not measure the full LLM divergence or probability outside that support. MSE-250 and WBC-250 reduce KL by 0.1675 and 0.1004, respectively; the reported intervals in Table 15 concern Pearson correlation, not KL improvement. Paired prompt intervals for KL remain to be reported. With uniform occurrence weights and $\Delta = 0$, target-argmax mass is algebraically 1/8; reference top-1 agreement depends on tie-breaking.

Implementation history. The repaired study changed target construction, reference handling, tokenization, sampling, and training data jointly. Earlier failures cannot identify the effect of any

Table 13: Target realization on 500 development prompts, one seed. Each horizon has its own cosine schedule; 250-update runs are development-informed follow-ups. Pool KL is $D_{\text{KL}}(p^* \parallel \hat{p}_\theta)$ on sampled support; the reference value is 0.7879. Full diagnostics are in Appendix E.

Projection	Horizon	nMSE	Sign	Pearson	Pool KL↓	GPU-h
MSE	1750	1.418	0.693	0.415	1.2131	7.48
WBC	1750	13.070	0.486	-0.152	6.0634	6.78
MSE†	250	0.690	0.702	0.588	0.6204	2.20
WBC†	250	0.780	0.687	0.492	0.6875	2.08

Table 14: All recorded development evaluations of the repaired runs. The first column identifies the declared schedule; “Updates” is the checkpoint step. Oracle nMSE* permits signed rescaling. No trajectory point is substituted for its run’s declared final checkpoint.

Projection	Updates	nMSE	nMSE*	Sign acc.	Pearson	Spearman	$\log(\pi_\theta / \pi_\tau)$
NBPO-MSE, 1750	250	0.839	0.734	0.692	0.516	0.499	-1.13
	500	1.226	0.796	0.688	0.452	0.465	-1.86
	750	1.482	0.824	0.678	0.419	0.452	-2.42
	1000	1.568	0.839	0.694	0.401	0.449	-2.38
	1250	1.517	0.836	0.688	0.404	0.450	-2.27
	1500	1.432	0.831	0.691	0.412	0.455	-2.12
	1750	1.418	0.828	0.693	0.415	0.460	-2.07
NBPO-WBC, 1750	250	0.920	0.824	0.645	0.419	0.415	-0.75
	500	1.932	0.938	0.623	0.250	0.297	-2.04
	750	3.776	0.995	0.569	0.074	0.124	-3.27
	1000	6.121	0.999	0.535	-0.028	0.018	-4.39
	1250	11.611	0.979	0.496	-0.143	-0.096	-6.15
	1500	12.672	0.979	0.489	-0.144	-0.097	-6.63
	1750	13.070	0.977	0.486	-0.152	-0.105	-6.68
NBPO-MSE, 250†	50	0.737	0.702	0.675	0.546	0.507	-0.77
	100	0.751	0.674	0.691	0.571	0.537	-1.11
	150	0.710	0.659	0.695	0.584	0.548	-1.03
	200	0.701	0.668	0.693	0.576	0.539	-0.87
	250	0.690	0.655	0.702	0.588	0.552	-0.90
NBPO-WBC, 250†	50	0.866	0.839	0.628	0.401	0.388	-0.39
	100	0.807	0.777	0.655	0.473	0.450	-0.55
	150	0.755	0.730	0.678	0.520	0.499	-0.64
	200	0.770	0.743	0.682	0.507	0.497	-0.57
	250	0.780	0.758	0.687	0.492	0.488	-0.51

single correction. Detailed debugging sweeps and the retired prompted-judge audit are retained in the code archive rather than used as performance evidence.

E.1 AGGREGATION-TARGET SENSITIVITY

On the reported pool, Nash weights (6.325, 6.186) yield targets close to equal weights with the same ℓ_1 mass. Mean target TV is 0.0008. Separately, pairwise-target difference RMS is 0.0034, approximately 0.30% of target RMS 1.125. These are different diagnostics and must not be conflated. The completed three-seed comparison does not resolve a policy advantage; it does not establish equivalence or infer statistical power from target distance.

F EVALUATION PROTOCOL

Teacher-defined and independent evaluation. SafeRLHF surplus uses the frozen training preference models on held-out prompts, common comparator pools, and the same definition of $\hat{d}$. Greedy responses define point-mass decoders; they are not unbiased samples of the trained stochastic policy. Their positive surplus does not certify Algorithm 1 acceptance. Any stochastic-policy evaluation must sample both the trained policy and reference under a common declared protocol and account for prompt uncertainty.

Table 15: Independent reconstruction on the development pool. Correlation intervals are prompt-clustered. KL improvement is reference minus policy KL. Target-argmax mass is $\hat{p}_\theta(i^*)$, $i^* \in \arg \max_i p^*(i)$; tie rules must be fixed. The WBC-1750 improvement is reported at source precision and differs by 10^{-4} from subtracting rounded KL entries.

Policy	Pearson [clustered CI]	$D_{KL} \downarrow$	KL improvement $\uparrow$	Top-1	Argmax mass
Base	—	0.7879	+0.0000	0.108	0.1250
MSE (1750)	+0.417 [0.370, 0.463]	1.2131	−0.4252	0.420	0.3135
WBC (1750)	−0.150 [−0.205, −0.095]	6.0634	−5.2754	0.124	0.1113
MSE (250) †	+0.589 [0.554, 0.622]	0.6204	+0.1675	0.406	0.2906
WBC (250) †	+0.496 [0.459, 0.531]	0.6875	+0.1004	0.362	0.2283

Table 16: Measured direct-solver scaling by prompt count, pool size, and CPU workers. Times cover finite-pool solving and artifact generation, not LLM rollout or training.

prompts	pool	workers	solve (s)	dual evals	s/eval	proj. KKT	inner resid.	peak RSS (MB)	artifact (s)
200	4+4	32	1.5	31	0.047	3×10^{-14}	3×10^{-11}	467	0.01
200	8+8	32	1.7	35	0.049	4×10^{-15}	5×10^{-12}	498	0.02
200	16+16	32	1.8	32	0.055	2×10^{-14}	10^{-11}	499	0.02
1000	4+4	96	4.4	33	0.132	7×10^{-15}	2×10^{-11}	499	0.03
1000	8+8	96	3.2	26	0.122	4×10^{-15}	6×10^{-12}	504	0.04
1000	16+16	96	4.4	37	0.119	3×10^{-13}	10×10^{-11}	534	0.07
7000	4+4	96	10.6	28	0.379	7×10^{-10}	4×10^{-11}	534	0.11
7000	8+8	96	10.9	27	0.404	2×10^{-15}	3×10^{-11}	544	0.22
7000	16+16	96	15.3	30	0.508	10^{-14}	7×10^{-11}	633	0.39

IFEval, GSM8K, harmful-completion checks, and benign-refusal evaluation measure complementary behavior. AlpacaEval and Arena-Hard prompt sets are scored by a local reward model; these are not official leaderboard scores. Table 20 preserves the execution report’s Arena aggregate and token medians, whose subset and prompt scope remain unspecified. They do not populate subset-specific main-table cells. The exact IFEval strict metric and numerical confidence intervals must be reconciled with evaluator outputs. A confidence interval containing 0.5 does not establish win-rate equivalence.

Refusal concentration and teacher truncation. The audit distinguishes identical-string responses (247/1000), short responses (450/1000), refusal-prefixed responses (621/1000), and refusals among short responses (448/450). These output-defined categories overlap and are not interchangeable. Gains in a refusal subgroup do not identify a Nash-specific mechanism.

Teacher truncation affects $26,790/84,000 = 31.89\%$ of candidate occurrences; prompt truncation affects one occurrence. Among truncated responses, the reported median loss is 232 tokens and the observed response fraction is 61%. The latter quantities must retain their aggregation definitions. This is a response-tail visibility issue for the implemented teacher. The planned instruction panel therefore uses a natively supported longer-context encoder and a separate full-response evaluator, rather than increasing RoBERTa’s length limit. Reported joint-positive rates 0.725/0.730 are archived with their unit unresolved; they are not population IR certificates.

Common evaluation and mechanism comparison. Within a panel, all policies use the same frozen evaluator, disagreement construction, comparator bank, and evaluation temperatures, regardless of training temperatures. The planned mechanism comparison reuses fresh generations from the reference, NBPO, fixed-reference Nash, and one development-selected competitor on a fixed held-out subset. Independently judged cross-play and common-temperature finite-bank game values complement direct-reference win rates. Comparator selection is frozen on development data; this comparison remains unmeasured.

Feedback budgets and provenance. We record unique semantic comparisons separately from order-swapped calls, generated tokens, preference-model fitting/scoring, solver time, neural training, and tuning trials. Numerical iterations reuse cached matrices and do not supply new labels. Budget-matched claims require a shared feedback contract and total-compute accounting. All responses, derived pairs, and reversed orders from one source prompt remain in the same split. Manifests identify feedback source, teacher/rubric revisions, objective order, split, comparator construction, and

Table 17: Nash versus mass-matched utilitarian target diagnostics. TV and relative pairwise-target RMS measure different quantities.

Diagnostic	Value
Mean target TV	0.0008
Maximum target TV	0.0075
Pairwise target difference RMS	0.0034
Target RMS on the aggregation panel	1.125
Nash λ_{help}	6.325
Nash λ_{safe}	6.186

Table 18: Paired Nash minus utilitarian surplus differences on the same 1,000 prompts, conditional on each policy seed. Reported intervals include zero; their exact construction and confidence level must be retained from the artifacts. They do not establish equivalence.

Seed	Δs_{help}	Reported CI	Δs_{safe}	Reported CI
42	-0.0003	[-0.0067, +0.0058]	-0.0010	[-0.0074, +0.0050]
43	+0.0019	[-0.0038, +0.0076]	+0.0004	[-0.0052, +0.0061]
44	-0.0033	[-0.0086, +0.0023]	-0.0037	[-0.0095, +0.0017]

decoding. Historical pair-split, precursor, or differently judged results are not pooled with the repaired panel. Instruction-following error types, length, and refusal patterns are descriptive diagnostics; output-defined subgroups alone do not identify a causal mechanism.

F.1 EMPIRICAL PARETO EVALUATION

The population Nash optimum is Pareto efficient under the stated assumptions; a single proximal step and approximate neural realization do not inherit this certificate. The planned instruction comparison evaluates all four attributes using a fixed independent local judge. A policy is empirically nondominated only relative to the evaluated policies and evaluator in this four-dimensional space. The two main-figure projections are not complete frontiers.

Scalarized DPO (Rafailov et al., 2023) uses scalar BT rewards calibrated on development data. The declared weight set contains the four coordinate vectors, (.25, .25, .25, .25), (.1, .1, .1, .7), and (.1, .7, .1, .1), in IF/TF/Hon/Help order. Its main-table row denotes the uniform setting; all weight settings appear in the trade-off data. Actual MOPO and PROSPER adaptations preserve their respective constraints and game updates. All rows are unmeasured until corresponding outputs are available.

Compare methods on the complete final set and prespecified label-disagreement subsets. Keep independent win rates separate from training-teacher game values. Report individual policy seeds, paired prompt uncertainty, and all declared weights. Curves joining points do not represent evaluated mixture policies. Nash welfare is defined only for positive game surpluses; mark other cases infeasible. Fix hypervolume normalization and reference points before final scores are opened.

Uncertainty and provenance. Report every policy seed, then mean and standard deviation across training seeds. Each paired prompt-bootstrap replicate recomputes objective means and then the minimum. Its intervals condition on a checkpoint and do not include training-seed variability. Freeze development choices before confirmatory comparisons and retain the history of benchmark exposure. Nash/utilitarian aggregates use three policy seeds, while the historical WBC/long-schedule snapshots use one. Missing method-wise values must not be reconstructed from ranges pooled across methods; dashes are unmeasured, not zeros. All result updates must record checkpoint, split, decoding, evaluator, sample count, and source artifact. Unmatched literature scores do not fill matched-baseline rows.

Table 19: Historical single-checkpoint objective evaluations. Short-run objective entries retain their reported precision. These rows do not replace the completed three-seed comparison in Table 3.

Method	Updates	$s_{\text{help}} \uparrow$	$s_{\text{safe}} \uparrow$	$\min_k s_k \uparrow$
Finite-pool target p^*	—	+0.1653	+0.1662	+0.1653
Base (greedy)	—	−0.0405	−0.0320	−0.0405
NBPO-MSE (1750)	1750	+0.1339	+0.1337	+0.1337
NBPO-WBC (1750)	1750	+0.0738	+0.0808	+0.0738
NBPO-MSE (250) [†]	250	+0.119	+0.125	+0.1186
NBPO-WBC (250) [†]	250	—	—	+0.0569
Game-utilitarian	250	—	—	—
Fixed-reference Nash	250	—	—	—
BT-RM–Nash	250	—	—	—
PROSPER	—	—	—	—
MOPO	—	—	—	—
Game-maxmin	250	—	—	—
Scalarized DPO	—	—	—	—

Table 20: Completed downstream snapshot supplied by the execution report. Arena is an unassigned local-proxy aggregate; token medians have an unassigned prompt scope. These values are preserved without relabeling them as hard/creative or Alpaca-specific results. This historical snapshot contains one checkpoint per row; it is not the new three-seed aggregate.

Policy	$\min_k s_k$	IFEval	GSM8K	Harm↓	Alpaca	Arena (unassigned)	Tokens (unassigned)
Base	−0.0405	.752	.867	.281	.500	.500	457
MSE (1750)	+0.1337	.758	.864	.284	.419	.435	481
WBC (1750)	+0.0738	.741	.852	.256	.578	.594	457
MSE (250) [†]	+0.1186	.760	.870	.269	.497	.516	460
WBC (250) [†]	+0.0569	.741	.867	.266	.486	.542	463

Table 21: Evaluation contracts and fields to complete from actual artifacts. These counts specify the requested full evaluation sets, not a claim that every job has completed.

Set / target count	Metric and completion contract
SafeRLHF / 1000	Teacher-defined objective values; comparator and disagreement hashes; greedy and stochastic evaluations kept separate.
IFEval / 541	Official strict prompt and instruction accuracy; loose scores secondary. Verify the reported strict column against evaluator output.
GSM8K / 1319	Fixed zero-shot-CoT prompt and answer extractor; exact match and parse-failure count.
HarmBench / text test	Report actual category counts and classifier version. Use category-specific hashing where required.
XSTest / 250 safe, 200 unsafe	Benign refusal and unsafe-request handling separately; report classifier or blinded manual assessment.
Alpaca / 805	Same-prompt local reward-model win rate versus base, ties half; response length from the same prompt set.
Arena / 500 hard, 250 creative	Report subset-specific local proxy win rates; an unassigned aggregate does not populate these columns.
MT-Bench / 80	Optional two-turn responses; no official score without the stated judging protocol.